

PAKOWKI LAKE AGCM, AB, Canada

WMO: 710700

Lat: 49.2247N

Lon: 111.1261W

Elev: 3002

StdP: 13.17

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.0	-14.7	-26.9	1.4	-20.8	-20.2	2.0	-14.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.2	90.7	60.3	87.2	60.2	83.5	59.5	65.2	78.9	63.3	78.9	61.6	77.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	88.3	70.7	58.1	80.7	67.5	56.1	74.9	65.5	31.8	77.6	30.2	78.7	28.9	77.7	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-29.9	96.7	4.5	3.0	-33.1	98.8	-35.7	100.6	-38.2	102.2	-41.5	104.4
			-30.1	68.6	4.4	3.1	-33.3	70.8	-35.9	72.7	-38.3	74.4	-41.5	76.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	19.1	17.7	30.5	41.6	52.0	59.8	66.3	65.4	56.4	42.9	30.1	17.6	(6)
	DBStd	20.94	15.02	15.84	13.75	9.18	7.97	6.12	5.34	6.40	8.71	9.75	13.87	15.85	(7)
	HDD50	4716	959	905	605	274	73	5	0	1	35	256	600	1003	(8)
	HDD65	8706	1424	1325	1068	703	407	177	48	76	280	684	1046	1468	(9)
	CDD50	1707	0	0	2	21	134	299	506	477	227	37	4	0	(10)
	CDD65	222	0	0	0	0	4	21	88	87	22	0	0	0	(11)
	CDH74	4271	0	0	0	17	173	424	1524	1574	538	21	0	0	(12)
	CDH80	1581	0	0	0	2	32	113	580	650	203	1	0	0	(13)
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
	PrecAvg	14.2	0.7	0.6	0.8	1.2	1.8	3.2	1.4	1.3	1.3	0.8	0.6	0.6	(15)
(16) Precipitation	PrecMax	21.7	1.9	1.4	1.8	2.7	3.5	8.2	5.0	3.6	2.7	2.7	1.2	1.3	(16)
	PrecMin	8.3	0.3	0.1	0.0	0.2	0.0	0.2	0.1	0.0	0.0	0.2	0.1	0.0	(17)
	PrecStd	3.2	0.4	0.3	0.5	0.6	1.0	1.6	1.1	0.9	0.8	0.6	0.3	0.4	(18)
	PrecStd	3.2	0.4	0.3	0.5	0.6	1.0	1.6	1.1	0.9	0.8	0.6	0.3	0.4	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.6	54.4	64.0	76.0	83.7	89.9	94.1	94.9	90.9	77.0	65.3	52.6	(19)
		MCWB	40.7	43.7	46.4	52.1	56.5	63.5	61.5	60.8	58.8	54.4	48.2	41.1	(20)
	2%	DB	45.1	46.8	58.8	69.0	79.1	82.9	90.3	90.9	86.4	69.6	57.1	45.8	(21)
		MCWB	37.6	39.1	44.3	49.8	55.4	59.9	61.3	59.7	58.3	51.1	44.9	38.2	(22)
	5%	DB	41.6	41.5	53.9	63.0	74.3	78.7	86.4	87.3	80.4	64.0	52.1	41.7	(23)
		MCWB	36.2	35.6	42.2	46.2	53.3	58.6	61.5	59.3	56.1	48.9	42.3	35.6	(24)
	10%	DB	38.8	38.2	48.5	58.4	69.5	74.9	82.7	83.2	74.6	59.0	47.7	38.4	(25)
		MCWB	34.7	33.7	39.0	43.7	51.7	56.8	60.9	59.2	54.0	46.2	39.5	33.7	(26)
	0.4%	WB	41.5	43.9	47.9	53.5	57.9	66.5	69.2	64.8	62.5	56.8	49.5	42.7	(27)
		MCDB	49.2	53.6	61.9	71.1	78.6	77.9	82.1	80.0	78.6	74.3	63.3	49.8	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.1	39.2	45.3	50.2	56.2	63.3	66.5	63.1	59.6	52.9	45.9	38.5	(29)
		MCDB	44.0	45.9	57.1	67.4	74.4	79.7	79.0	80.6	79.7	65.5	55.6	44.7	(30)
	5%	WB	36.4	35.9	42.6	47.2	54.5	60.6	64.5	61.7	57.4	49.8	43.0	36.0	(31)
		MCDB	41.1	40.8	52.7	61.0	70.5	74.4	78.5	80.2	77.1	61.4	51.5	41.0	(32)
	10%	WB	34.6	33.8	39.6	44.4	52.8	58.5	62.6	60.3	55.3	47.2	40.0	33.8	(33)
		MCDB	38.8	37.9	47.9	56.4	66.6	71.1	77.6	78.5	73.3	58.1	46.9	38.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	19.8	20.7	23.1	23.5	23.7	28.2	28.8	26.8	23.3	20.5	18.5	(35)
		MCDBR	19.0	22.1	27.3	33.0	32.7	30.9	35.3	37.0	37.0	30.7	25.3	20.0	(36)
		MCWBR	13.7	15.3	15.2	16.4	13.3	12.8	13.4	12.9	14.6	15.2	14.7	13.8	(37)
	5% WB	MCDBR	18.4	21.4	26.2	31.4	28.1	27.7	28.7	30.4	34.1	28.5	24.5	20.1	(38)
		MCWBR	13.7	15.4	15.4	16.1	12.3	12.9	13.2	11.6	14.0	14.8	15.0	14.6	(39)
			(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			(46)
(42) Clear-Sky Solar Irradiance	taub		0.260	0.265	0.286	0.335	0.352	0.347	0.347	0.368	0.326	0.291	0.261	0.250	(40)
	taud		2.440	2.483	2.487	2.351	2.346	2.414	2.425	2.338	2.429	2.518	2.505	2.432	(41)
	Ebn at Noon		260	287	298	290	287	289	287	275	278	273	259	248	(42)
	Edh at Noon		19	23	28	36	38	36	35	36	30	23	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg		409	718	1128	1562	1856	2071	2253	1813	1291	809	461	320	(44)
	RadStd		34	60	85	101	140	121	116	113	102	77	31	28	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page